

AI-Based ABADS + SENTINEL Defence System

A Complete Proposal for Battlefield Awareness, Mine Detection and Acoustic Direction System

Abstract

This project presents an integrated battlefield system combining ABADS and SENTINEL modules for threat detection, mapping, and decision support without reliance on GPS or internet.

Introduction

Modern warfare demands intelligent systems capable of autonomous sensing, analysis, and decision support. This project addresses these needs.

Problem Statement

Soldiers face risks due to landmines, unclear threat direction, and communication challenges in GPS-denied and jammed environments.

Objectives

To design a system that detects threats, maps risks, and provides actionable guidance without external dependencies.

System Overview

The system integrates UGV-based sensing with fixed acoustic nodes to provide full battlefield awareness.

ABADS Module

Detects mines, maps terrain risk, and suggests safe paths using AI and sensor fusion.

SENTINEL Module

Uses multiple microphones to detect sudden acoustic events and estimate direction using time difference of arrival.

Acoustic Theory

When a sound occurs, it reaches microphones at different times. This difference is used to calculate direction.

System Architecture

UGV + Sensor nodes → Local processing → Command unit → Decision output

Communication Without Internet

Uses RF modules, mesh networking, and local processing. No dependency on internet or GPS.

Anti-Jamming Techniques

Frequency hopping, local AI processing, minimal communication, and redundancy systems.

Fixed vs Mobile Deployment

System can be deployed as fixed forward post sensors or mobile UGV. Hybrid approach recommended.

Range Analysis

Prototype acoustic range: 10–30m. RF range: 100–500m. Military systems extend this significantly.

Components

Microphones, Raspberry Pi, Arduino, sensors, RF modules, power systems.

Implementation Plan

Phase 1: Hardware setup. Phase 2: Sensor integration. Phase 3: AI processing. Phase 4: Testing.

Applications

Military mine detection, forward surveillance, disaster rescue, industrial safety.

Advantages

Reduces human risk, autonomous operation, works in denied environments.

Limitations

Prototype accuracy limited, environmental noise impacts performance.

Future Scope

Integration with drones, thermal sensors, and advanced AI models.

Conclusion

This project demonstrates a scalable, intelligent battlefield system aligning with modern defence needs.